

EXECUTIVE SUMMARY

Predicting Loan Default Risk for a Financial Institution

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ADTA 5340 – Discovery and Learning with Big Data

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This project applies Kaggle's Loan Default Prediction dataset to address the challenge of predicting loan default risk for a financial institution. The goal is to develop a strong classification model which can identify high-risk borrowers prior to loan approval, helping the organization in reducing losses while improving risk-based decision-making. The dataset presents a thorough foundation for predictive modeling that incorporates borrower demographics, employment details, financial ratios, credit history, and loan characteristics.

The data preparation process includes handling missing values, addressing outliers, encoding categorical variables, and engineering key features such as high debt-to-income (DTI) indicators. The exploratory analysis observed clear trends: borrowers with greater DTI ratios and weaker credit profiles showed much greater default rates. These insights improved feature selection and presented an initial understanding of risk distribution among applicant segments.

The significant difference between default and non-default situations was the primary challenge in the dataset. To address this, the training data went through the Synthetic Minority Oversampling Technique (SMOTE), which improved the model's ability to detect minority-class (default) patterns. Several machine learning algorithms were evaluated, including Logistic Regression, Random Forest, and XGBoost. Models were evaluated using accuracy, precision, recall, F1-score, and AUC in order to provide an accurate evaluation of predictive performance.

Among all the models The XGBoost classifier trained with SMOTE, using a decision threshold of 0.3 was the best-performing model. With an AUC of 0.75 and a recall of 88.6% for default cases, this setup is an ideal choice for real-world lending applications. Although the lower threshold increases false positives, it significantly lowers the probability of missing true defaulters, which is consistent with the institution's goal to minimize loan losses. Feature importance analysis highlighted that the DTI ratio, income level, and credit history variables were the most effective predictors of default risk.

Based on the model's findings, the project suggests more screening for high-DTI applicants, using revised loan terms for moderate-risk borrowers, and incorporating the predictive model into the loan approval workflow. By proactively identifying high-risk individuals and allowing data-driven lending strategies, such measures can significantly improve portfolio quality. The final model indicates how machine learning may help with financial risk management by enhancing early detection of potential defaults.

Reference:

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